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SEPM Assignment - 1

Q1. Define a Project? What is the project management? Explain various characteristics of project management?

→ A project is a temporary endeavor undertaken to create a unique product, service, or result. It has a defined beginning and end and it is usually constrained by funding, resources and time.

Project management.

→ project management is the application of knowledge, skills, tools and techniques to project activities to meet the project requirements. It involves planning, executing and monitoring tasks to ensure the goal is achieved within the set constraints.

Characteristics of project management.

→ Temporary Nature: Every project has a specific start date and a specific end date once the objectives are met.

→ Unique Deliverables: Projects aim to produce something that hasn't been done exactly that way before, distinguishing it from routine operations.

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- Progressive Elaboration: Projects are often defined broadly at the start and become more detailed as the team gains a better understanding of the objectives.
 - Resource constraints: management must balance limited resources including budget, manpower and equipment.
 - Defined objectives: success is measured by how well the project meets its pre defined goals and quality must be managed. Standards
 - Risk and Uncertainty: Because projects are unique, they inherently involve a degree of risk that must be managed.
 - Cross - functional Teams: Projects often require a diverse group of people from different departments or specialities to work together.

Q2: Write a short note on PMBOK

- The project management Body of knowledge (PMBOK) is a set of standard terminology and guidelines for project management.

- Global standard: maintained by the project management Institute (PMI), it is recognized worldwide as the "bible" of project management.
- Process - Based: It is described project management in terms of processes, which overlap and interact throughout a project's life cycle.
- Process groups: It categorizes project activities into five groups: initiating, planning, executing, monitoring & controlling and closing.
- Knowledge areas: It defines 10 knowledge areas such as scope, schedule, cost, quality and Risk management.
- Best practices: It provides a framework of proven traditional practices that are widely applied as well as innovative emerging practices.
- Professional Certification: It serves as the primary fundamental resource for the project management Professional (PMP) certification.
- Adaptability: While comprehensive, the PMBOK emphasizes "tailoring" meaning managers should use the portions of the guide appropriate for their specific project.

Q3 Explain ITPM?

- Information Technology Project Management (ITPM) is the process of managing the plan, organization and accountability to achieve specific information technology goals.
- IT focus: Unlike general project management, ITPM deals specifically with IT infrastructure software development or systems integration.
- Complex life cycles: often involves complex software development life cycles (SDLC) like waterfall, agile or scrum.
- Technical stakeholders: requires managing communication between technical teams (developers, IT ops) and non-technical business stakeholders.
- High change rate: ITPM must account for rapid changes in technology and scope creep which are common in software projects.
- Integration requirements: ITPM often focuses heavily on how new systems will integrate with existing legacy hardware and software.

→ Data security: A critical component of ITPM is ensuring that the project results comply with data privacy and cybersecurity standards.

→ Testing and QA: Significant emphasis is placed on rigorous testing phases (Unit integration, UAT) to ensure the IT solution is functional.

Q4 What is the role of RFP? Describe its content and purpose?

→ A Request for Proposal (RFP) is a document that an organization posts to elicit bids from potential vendors for a product or service.

Purpose of RFP

→ solicit competitive Bids: To get the best value by comparing different vendors solutions and pricing

→ Define requirements: clearly communicates what the organization needs so vendors can provide accurate solutions.

→ Transparency: Provides a fair and structured environment for the procurement process.

Content of an RFP

→ Executive Summary: A brief overview of the project and its goals.

- Company background: information about the organization to help vendors understand the business context
- Project scope and specifications: Detailed technical and functional requirements of the desired solution
- Submission guidelines: Deadline, formats and contact information for the proposal.
- Evaluation Criteria: How the organization will score or judge the proposals
- Timeline: The expected schedule for the selection process and the project itself
- Budget Constraints: sometimes included to ensure vendors propose solutions within a specific financial range

Q5 What is feasibility study? Explain its types, contents and purpose?

- A Feasibility study is an assessment of the practicality of a proposed project or system

Purpose

- Decision making: To determine whether the project should move forward or be abandoned.
- Risk Identification: To uncover potential legal, technical or financial bottlenecks before investing.

Types of feasibility

- Technical Feasibility: Can we build it? Does the technology exist?
- Economic Feasibility: Is it worth it? (cost benefit analysis).
- Legal Feasibility: Does it comply with laws and regulations (GDPR, labor laws, etc)?
- Operational Feasibility: Will it work in our current organizational environment?
- Scheduling Feasibility: Can it be completed in a reasonable amount of time?

Contents of a Feasibility report

- Description of the Project: Overview of the intended product or service.

- market analysis: Research on the demand or need for the project.
- Technical requirements: analysis of hardware, software, and skilled labour needed.
- Financial projections: Estimates of costs and expected return on Investment (ROI).
- ~~Scheduling~~ Scheduling
- Risk Assessment: Identification of major risks and mitigation strategies.

Q6 Write a short note on W.B.S

- The work Breakdown Structure (WBS) is a hierarchical decomposition of the total scope of work to be carried out by the project team.
- Visual hierarchy: It breaks a large project down into smaller, manageable pieces called "work packages".
- Deliverable - oriented: Focuses on what needs to be delivered rather than just the actions taken.
- 100% Rule: The WBS should include 100% of the work defined by the project scope and nothing ^{more}.

- Estimation Basis: Provides a framework for detailed cost estimation and scheduling.
- Accountability: Helps assign specific responsibilities to individuals or teams for each work package.
- Progress Tracking: Makes it easier to monitor which parts of the project are finished and which are lagging.
- Communication Tool: Acts as a common reference point for stakeholders to understand the project's structure.

Q7 Consider a XYZ company undertakes a project to computerized working of ABC City Bank, Then - Develop W.B.S

→ For the project of computerizing a city bank the Work Breakdown Structure (WBS) can be organized into the following levels:

1. Project Management
 - 1.1 Planning & sp. scoping
 - 1.2 Resource Allocation
 - 1.3 stakeholder meetings.

2. Infrastructure & Hardware

2.1 server installation

2.2 Networking & connectivity (LAN/WAN)

2.3 security Systems (Biometrics/CTV)

3. software development / customization

3.1 Database Design (customer & Transaction Records)

3.2 Core Banking Module (savings, current loans)

3.3 User Interface (UI) for Tellers.

4. Data Migration.

4.1 Digitizing physical records

4.2 Data cleaning & validation

4.3 Uploading to new Database

5. Testing & quality assurance

5.1 System Integration Testing

5.2 Security & Penetration Testing

5.3 User Acceptance Testing (UAT)

6. Deployment & Training

6.1 Staff Training Program

6.2 User Manual Documentation

6.3 Final "Go-Live" Transition